

CORPORATION SIMULATION

WAFT | Economic Modeling System

Overview

WAFT includes a full economic simulation system for modeling corporations. Track finances, employees, departments, and growth over time.

Corporation Structure

```
@dataclass
class Corporation:
    corp_id: str          # Unique identifier
    name: str             # Display name
    sector: str           # Industry sector
    mission: str          # Mission statement
    founded_date: datetime # Founding date
    capital: Decimal      # Current capital
    employees: dict       # Being ID → Employees record
    departments: list     # Department names
    monthly_expenses: list # Recurring costs
```

Creating Corporations

```
from waft.core import CorporationsSystem
from decimal import Decimal
from datetime import datetime

corps = CorporationsSystem(project_path=Path("."))

corp = corps.create_corporation(
    name="My Startup",
    sector="AI Research",
    mission="Build intelligent systems",
    founded_date=datetime(2026, 1, 1),
    initial_capital=Decimal("500000"),
)
```

Financial Tracking

Revenue & Expenses

```
# Add revenue
corp.add_revenue(
    amount=Decimal("50000"),
    source="Contract work",
    date=datetime.now(),
)

# Add expense
corp.add_expense(
    amount=Decimal("15000"),
    category="Equipment",
    description="Quantum analyzer",
    vendor="QuantumTech Inc",
)
```

Monthly Recurring Expenses

```
simulator.add_monthly_expense(
    description="Office rent",
    amount=Decimal("10000"),
    category="rent",
    vendor="Building Corp",
)
```

Balance Sheet

Item	Debit	Credit
Initial Capital		\$500,000
Equipment Purchase	\$15,000	
Contract Revenue		\$50,000
Monthly Rent	\$10,000	
Balance		\$525,000

Employee Management

Hiring

```
corp.hire_employee(
    being_id=being.being_id,
    role="Engineer",
    department="R&D",
    title="Senior Engineer",
    level=7,
    salary=Decimal("120000"),
)
```

Department Structure

```
corp.add_department("Executive")
corp.add_department("Research & Development")
```

```
corp.add_department("Operations")
corp.add_department("Finance")
```

Time Simulation

```
from waft.core.simulation import CorporationSimulator, TimeUnit
```

```
simulator = CorporationSimulator(
    corporation=corp,
    time_unit=TimeUnit.DAILY,
    start_date=datetime(2026, 1, 1),
)
```

```
# Advance time
simulator.advance(days=30)
```

```
# Check finances
print(f"Capital: ${corp.capital}")
print(f"Burn rate: ${simulator.monthly_burn_rate}")
print(f"Runway: {simulator.runway_months} months")
```

Reports

```
# Generate financial report
waft corp report my_startup --type financial
```

```
# Employee roster
waft corp roster my_startup
```

```
# Department summary
waft corp departments my_startup
```

CORPORATION SIMULATION | Business Evolution